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Aptean Merlin Edition

Merlin
Web Services Toolkit

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Introduction

The Merlin Web Module is an add-on designed to help you build websites around the Merlin Business System. It provides a set of functions and routines that enable the easy extraction of data for use and manipulation in an external website/application, as well the ability to generate new orders and accounts within Merlin.

We offer two different ways of providing this information, through an XML Web Service and using XML File Transfers.

Merlin Web Service

The Merlin Web Service is an ASP.NET Web Service that provides a number of functions to interact with and extract data from within the Merlin system. The web service benefits from having a constant link to the Merlin system enabling instant access to current data. The web service can be called from a website whenever a piece of information is required using a variety of coding languages (including ASP.NET and PHP).

At present, the Merlin Web Service accepts the following request methods: -

- Soap 1.1
- Soap 1.2
- HTTP POST
- HTTP GET

As the service can look at current data, it is necessary to provide a *datasource* parameter, for all functions that are used to control access to data. This information will be provided by Merlin upon request and must be kept secure by the developer.

Version

You can check the current version of the Web Service by navigating to the Merlin.asmx page within your web server (i.e. http://localhost:8181/Merlin.asmx).

CashReceipt

The `CashReceipt` function provides the ability to pay off an Invoice held on Merlin. A list of transactions id's can be passed through with a payment amount which will be 'paid' against the corresponding transaction in Merlin.

Parameters

Parameter	Type	Notes
Datasource	string	Connection string provided by Merlin
Xml	string	Validated against CashReceipt.xsd

XML Parameter XSD

A link to the XSD for the XML parameter of the `CashReceipt` function can be found at:

<<YOUR-WEB-SERVICE-URL>>/Schemas/CashReceipt.xsd

Sample XML Parameter

```
<cashreceipt>
  <customerid>1234</customerid>
  <reference>ABCDEFGH</reference>
  <transaction>
    <transactionid>123456</transactionid>
    <payment>110.00</payment>
```


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```
    <settlement>10</settlement>
  </transaction>
  <transaction>
    <transactionid>123457</transactionid>
    <payment>100.00</payment>
  </transaction>
</cashreceipt>
```

Sample Result

```
<status>
  <code>1</code>
  <message></message>
</status>
```

Upon successful processing of an order, the web service will return a status message containing:

Code = 1
Message = *empty*

CheckStatus

The `CheckStatus` function allows you to verify that the web service is successfully connected to Merlin using the data source you specify.

Parameters

Parameter	Type	Notes
datasource	string	Connection string provided by Merlin

Sample Result

```
<status>
  <code>0</code>
  <message>Connected</message>
</status>
```

Upon successful submission, the web service will return a status message containing:

Code = 0
Message = Connected

CustomerPoints

The `CustomerPoints` function provides the ability to update a customer's sales points.

Parameters

Parameter	Type	Notes
Datasource	string	Connection string provided by Merlin
Customerid	integer	A valid customer ID
ref1	string	A reference for these points
Earned	integer	The number of points earned
Redeemed	integer	The number of points redeemed

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Sample Result

```
<status>
  <code>0</code>
  <message>OK</message>
</status>
```

Upon successful processing of an order, the web service will return a status message containing:

Code = 0
Message = *OK*

GetSql

The `GetSql` function allows an SQL query to be executed on the Merlin database and the result returned as an xml document to the user. This can be used to obtain specific data from Merlin that may not be possible with any other existing functions. The function only allows `SELECT` statements to be run, preventing users from manipulating data within Merlin.

Parameters

Name	Type	Notes
datasource	string	Connection string provided by Merlin
query	string	A valid SQL select query

Sample Query Parameter

```
SELECT customer.account, customer.country
FROM customer
WHERE customer.company = 1
ORDER BY customer.account
LIMIT 3
```

Sample Result

```
<data count="3" count_available="1050">
  <row id="1">
    <account>ABC123</account>
    <country>England</country>
  </row>
  <row id="2">
    <account>ENG002</account>
    <country>England</country>
  </row>
  <row id="3">
    <account>SPA001</account>
    <country>Spain</country>
  </row>
</data>
```

GetStock

The `GetStock` routine is used to return a customer's special price and available quantity for a single stock part dependent on the quantity entered. It calculates a price based on the same routines used within Merlin to ensure that prices will be consistent.

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Parameters

Parameter	Type	Notes
datasource	string	Connection string provided by Merlin
company	integer	A valid company number
account	string	A valid customer account
stockid	integer	A valid stock id
quantity	decimal	Quantity of stock required

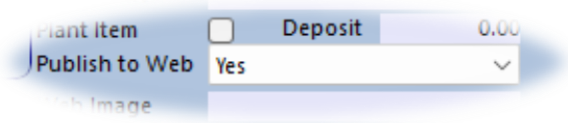
Sample Result

```
<stock>
  <stockid>9</stockid>
  <part>abc</part>
  <desc>abc</desc>
  <price>9.99</price>
  <disc1>9.99</disc1>
  <disc2>9.99</disc2>
  <disc3>9.99</disc3>
  <std_price>9.99</std_price>
  <disc_price>9.99</disc_price>
  <qty_free>9</qty_free>
</stock>
```

Upon successful submission, the web service will return an XML document with root element `<stock>`.

GetStockList

The `GetStockList` routine is used to return a list of stock parts based on a given search criteria and where the web_available flag is set on the stock item.



Parameters

Parameter	Type	Notes
datasource	string	Connection string provided by Merlin
fields	string	List of specific fields to return (or empty for defaults)
search	string	Value to search for
start	integer	Offset starting position (good for paging)
numberresults	integer	Maximum number of stock records to return

Sample Result

```
<data count="1" count_available="9">
  <row id="1">
    <stockid>123</stockid>
    <company>1</company>
    <depot>01</depot>
    <part>ABC</part>
    <qty_free>123</qty_free>
```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```
    <desc1>abc</desc1>
    <desc2>abc</desc2>
    <desc3>abc</desc3>
    <desc4>abc</desc4>
    <desc5>abc</desc5>
    <list_price>11.11</list_price>
    <uoi>1</uoi>
    <duoi>EACH</duoi>
    <web_image>/images/abc.png</web_image>
    <wines_flag>0</wines_flag>
    <wines_abv>0</wines_abv>
    <wines_vintage>0</wines_vintage>
    <wines_region>abc</wines_region>
    <wines_volume>0</wines_volume>
  </row>
</data>
```


GetStockMulti

The `GetStockMulti` routine is used to return an array of prices, discounts, description and free quantity (based upon the requested quantities).

Parameters

Parameter	Type	Notes
datasource	string	Connection string provided by Merlin
company	integer	A valid company number
account	string	A valid customer account
input	decimal	Comma separated list of <stockid>,<quantity>
Rate	string	Currency Rate (if blank will be 1)

Sample Result


```
<data count="2">
    <row id="1">
        <stockid>166519</stockid>
        <part>B001</part>
        <desc>depot 01</desc>
        <price>350.00</price>
        <disc1>0.00</disc1>
        <disc2>0.00</disc2>
        <disc3>0.00</disc3>
        <std_price>410.00</std_price>
        <disc_price>350.00</disc_price>
        <qty_free>13992.32</qty_free>
        <pgroupid>0</pgroupid>
        <qty_hand>48468.320000</qty_hand>
        <lead_time>100</lead_time>
    </row>
    <row id="2">
        <stockid>166519</stockid>
        <part>B001</part>
        <desc>depot 01</desc>
        <price>300.00</price>
        <disc1>0.00</disc1>
        <disc2>0.00</disc2>
        <disc3>0.00</disc3>
        <std_price>410.00</std_price>
        <disc_price>300.00</disc_price>
```


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```
        <qty_free>13992.32</qty_free>
        <pgroupid>0</pgroupid>
        <qty_hand>48468.320000</qty_hand>
        <lead_time>100</lead_time>
    </row>
</data>
```

SetConsingment

The `SetConsignment` function is used to set the Consignment number on a Sales Order. It will also update the consignment number on any Dispatch Notes and Invoices for the Sales Order.

The consignment number will only be updated if the consignment number doesn't already exist for the order.

Parameters

Name	Type	Notes
datasource	string	Connection string provided by Merlin
xml	xml	Validated against SetConsignment.xsd

XML Parameter XSD

A link to the XSD for the XML parameter of the `SetConsignment` function can be found at:

<<YOUR-WEB-SERVICE-URL>>/Schemas/SetConsingment.xsd

Sample XML Parameter

```
<consignment>
  <company>1</company>
  <sales_order>S111111</sales_order>
  <consignment_number>abc</consignment_number>
</consignment>
```

Sample Result

```
<status>
  <code>0</code>
  <message>OK</message>
</status>
```

Upon successful submission, the web service will return a status message containing:

Code = 0
Message = OK

SetContact

The `SetContact` function is used to either create or edit a customer contact within Merlin. If the function is submitted with an XML containing an empty or zero `<keyfield>` element, then a new contact will be created for the customer with the specified `<customer_keyfield>`. If it is submitted with a value in the `<keyfield>` element, then the contact will be updated (if it exists within Merlin) or will be created using the specified account.

Parameters

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Parameter	Type	Notes
Datasource	string	Connection string provided by Merlin
Xml	string	Validated against SetContact.xsd

XML Parameter XSD

A link to the XSD for the XML parameter of the `SetContact` function can be found at:

<<YOUR-WEB-SERVICE-URL>>/Schemas/SetContact.xsd

Sample XML Parameter

```
<contact>
  <keyfield>0</keyfield>
  <company>1</company>
  <account>ABC</account>
  <customer_keyfield>0</customer_keyfield>
  <forename>ABC</forename>
  <middlename>ABC</middlename>
  <surname>ABC</surname>
  <salutation>ABC</salutation>
  <title>ABC</title>
  <add1>ABC</add1>
  <add2>ABC</add2>
  <city>ABC</city>
  <county>ABC</county>
  <country>ABC</country>
  <postcode>ABC</postcode>
  <office_phone>ABC</office_phone>
  <personal_phone>ABC</personal_phone>
  <mobile_phone>ABC</mobile_phone>
  <fax>ABC</fax>
  <sms>ABC</sms>
  <email>ABC</email>
  <memo>ABC</memo>
  <jobtitle>ABC</jobtitle>
  <accept_office>false</accept_office>
  <accept_personal>false</accept_personal>
  <accept_mobile>false</accept_mobile>
  <accept_fax>false</accept_fax>
  <accept_sms>false</accept_sms>
  <accept_email>false</accept_email>
  <default_contact>0</default_contact>
  <first_contact>1899-12-30 00:00</first_contact>
  <next_contact>1899-12-30 00:00</next_contact>
  <last_contact>1899-12-30 00:00</last_contact>
  <frequency>0</frequency>
  <default_filter>ABC</default_filter>
  <default_calltime>1899-12-30 00:00</default_calltime>
  <default_callday>0</default_callday>
  <department>ABC</department>
  <photograph>ABC</photograph>
  <operator>0</operator>
  <status>ABC</status>
  <default_op_contact>0</default_op_contact>
  <default_account_contact>0</default_account_contact>
  <disable_callman>false</disable_callman>
  <card_name>ABC</card_name>
  <card_token>ABC</card_token>
```


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```
  <card_lastfour>ABC</card_lastfour>
  <card_options>ABC</card_options>
  <userchk1>false</userchk1>
  <userchk2>false</userchk2>
  <userchk3>false</userchk3>
  <userchk4>false</userchk4>
  <userchk5>false</userchk5>
  <userchk6>false</userchk6>
  <userchk7>false</userchk7>
  <userchk8>false</userchk8>
  <userchk9>false</userchk9>
  <userchk10>false</userchk10>
  <userchk11>false</userchk11>
  <userchk12>false</userchk12>
  <userchk13>false</userchk13>
  <userchk14>false</userchk14>
  <userchk15>false</userchk15>
  <userchk16>false</userchk16>
  <userchk17>false</userchk17>
  <userchk18>false</userchk18>
  <userchk19>false</userchk19>
  <userchk20>false</userchk20>
  <userchk21>false</userchk21>
  <flag_override>0</flag_override>
  <user1>ABC</user1>
  <user2>ABC</user2>
  <user3>ABC</user3>
  <user4>ABC</user4>
  <user5>ABC</user5>
  <user6>ABC</user6>
  <user7>ABC</user7>
  <user8>ABC</user8>
  <user9>ABC</user9>
  <user10>ABC</user10>
  <userc1>ABC</userc1>
  <userc2>ABC</userc2>
  <userc3>ABC</userc3>
  <userc4>ABC</userc4>
  <userc5>ABC</userc5>
  <userc6>ABC</userc6>
  <userc7>ABC</userc7>
  <userc8>ABC</userc8>
  <userc9>ABC</userc9>
  <userc10>ABC</userc10>
  <usern1>0</usern1>
  <usern2>0</usern2>
  <usern3>0</usern3>
  <usern4>0</usern4>
  <usern5>0</usern5>
  <usern6>0</usern6>
  <usern7>0</usern7>
  <usern8>0</usern8>
  <usern9>0</usern9>
  <usern10>0</usern10>
  <alert_type>0</alert_type>
  <personal_folder>ABC</personal_folder>
  <gdpr_templateid>0</gdpr_templateid>
</contact>
```


Sample Result

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```
<status>
  <code>0</code>
  <message>123</message>
</status>
```

Upon successfully inserting a contact note, the web service will return a status message containing:

Code = 0
Message = *the generated customer contact ID*

SetContactNote

Available in Merlin v126 onwards.

The `SetContactNote` function provides the ability to create a new contact note for customer contacts.

Parameters

Parameter	Type	Notes
datasource	string	Connection string provided by Merlin
xml	string	Validated against SetContactNote.xsd

XML Parameter XSD

A link to the XSD for the XML parameter of the `SetContactNote` function can be found at:

<<YOUR-WEB-SERVICE-URL>>/Schemas/SetContactNote.xsd

Sample XML Parameter

```
<ContactNote>
  <company>1</company>
  <account>ABC</account>
  <customer_keyfield>1</customer_keyfield>
  <slcname_keyfield>1</slcname_keyfield>
  <note_date>1899-12-30 00:00</note_date>
  <note_note>ABC</note_note>
  <note_type>A</note_type>
  <filter>ABC</filter>
  <reference>ABC</reference>
  <operator>1</operator>
  <crm_data>ABC</crm_data>
  <prospect>0</prospect>
</ContactNote>
```

Sample Result

```
<status>
  <code>0</code>
  <message>55</message>
</status>
```

Upon successfully inserting a contact note, the web service will return a status message containing:

Code = 0
Message = *the generated contact note ID*

SetCustomer

Available in Merlin v100 onwards.

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The `SetCustomer` function is used to either create or edit a customer account within Merlin. If the function is submitted with an XML containing an empty `<account>` element, then a new customer will be created with an automatically generated account. If it is submitted with a value in the `<account>` element, then the account will be updated (if it exists within Merlin) or will be created using the specified account. This function will also set up delivery addresses for the customer if specified in the input xml.

Parameters

Name	Type	Notes
Datasource	string	Connection string provided by Merlin
Xml	xml	Validated against SetCustomer.xsd

XML Parameter XSD

A link to the XSD for the XML parameter of the `SetCustomer` function can be found at:

<<YOUR-WEB-SERVICE-URL>>/Schemas/SetCustomer.xsd

Sample XML Parameter

```
<customer>
  <company>9</company>
  <account>abc</account>
  <name>abc</name>
  <add1>abc</add1>
  <add2>abc</add2>
  <city>abc</city>
  <county>abc</county>
  <country>abc</country>
  <postcode>abc</postcode>
  <telephone>abc</telephone>
  <fax>abc</fax>
  <email>abc</email>
  <www>abc</www>
  <salesman>9</salesman>
  <statement_group>abc</statement_group>
  <user_1>abc</user_1>
  <user_2>abc</user_2>
  <user_3>abc</user_3>
  <invoiceaccount>MBS001</invoiceaccount>
  <statementaccount>MBS001</statementaccount>
  <depot>01</depot>
  <country_code>44</country_code>
  <back_orders>0</back_orders>
  <vat>H</vat>
  <user4>abc</user4>
  <user5>xyz</user5>
  <user6>123</user6>
  <user7>abc</user7>
  <user8>xyz</user8>
  <user9>123</user9>
  <user10>abc</user10>
  <userc1>abc</userc1>
  <userc2>xyz</userc2>
  <userc3>123</userc3>
  <userc4>abc</userc4>
  <userc5>xyz</userc5>
  <userc6>123</userc6>
  <userc7>abc</userc7>
  <userc8>xyz</userc8>
```

```
<userc9>123</userc9>
<userc10>abc</userc10>
<usern1>1</usern1>
<usern2>2</usern2>
<usern3>3</usern3>
<usern4>4</usern4>
<usern5>5</usern5>
<usern6>6</usern6>
<usern7>7</usern7>
<usern8>8</usern8>
<usern9>9</usern9>
<usern10>10</usern10>
<userchk1>false</userchk1>
<userchk2>true</userchk2>
<userchk3>false</userchk3>
<userchk4>true</userchk4>
<userchk5>false</userchk5>
<userchk6>true</userchk6>
<userchk7>false</userchk7>
<userchk8>true</userchk8>
<userchk9>false</userchk9>
<userchk10>true</userchk10>
<delivery_code>1</delivery_code>
<flagvat>0</flagvat>
<terms_code>1</terms_code>
<invoice_device>invoices@merlinsw.co.uk</invoice_device>
<statement_device>statements@merlinsw.co.uk</statement_device>
<nominal>12345</nominal>
<price_code>1</price_code>
<vatreg>GB</vatreg>
<account_type>A</account_type>
<statement_flag>2</statement_flag>
<co_reg>CRN</co_reg>
<delivery>
  <name>abc</name>
  <add1>abc</add1>
  <add2>abc</add2>
  <city>abc</city>
  <county>abc</county>
  <country>abc</country>
  <postcode>abc</postcode>
  <telephone>abc</telephone>
  <fax>abc</fax>
  <email>abc</email>
  <www>abc</www>
  <vat>H</vat>
  <flagvat>0</flagvat>
</delivery>
<delivery>
  <name>abc</name>
  <add1>abc</add1>
  <add2>abc</add2>
  <city>abc</city>
  <county>abc</county>
  <country>abc</country>
  <postcode>abc</postcode>
  <telephone>abc</telephone>
  <fax>abc</fax>
  <email>abc</email>
  <www>abc</www>
```


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```
    <vat>H</vat>
    <flagvat>0</flagvat>
  </delivery>
</customer>
```

Sample Result

```
<status>
  <code>0</code>
  <message>WEB0000000099</message>
</status>
```

Upon successful submission, the web service will return a status message containing:

Code = 0
Message = the generated/edited customer account

SetDelivery

Available in Merlin v129 onwards.

The `SetDelivery` function is used to either create or edit a customer delivery address within Merlin. If the function is submitted with an XML containing an empty `<deliveryid>` element, then a new delivery address will be created for the customer with the specified `<customerid>`. If it is submitted with a value in the `< deliveryid >` element, then the delivery address will be updated (if it exists within Merlin) or will be created using the specified account.

Parameters

Name	Type	Notes
datasource	string	Connection string provided by Merlin
xml	xml	Validated against SetDelivery.xsd

XML Parameter XSD

A link to the XSD for the XML parameter of the `SetDelivery` function can be found at:

<<YOUR-WEB-SERVICE-URL>>/Schemas/SetDelivery.xsd

Sample XML Parameter

```
<delivery>
  <customerid>1</customerid>
  <name>abc</name>
  <add1>abc</add1>
  <add2>abc</add2>
  <city>abc</city>
  <county>abc</county>
  <country>abc</country>
  <postcode>abc</postcode>
  <telephone>abc</telephone>
  <fax>abc</fax>
  <email>abc</email>
  <www>abc</www>
  <salesman>ABC</salesman>
  <vat>H</vat>
  <area>abc</area>
  <defaultaddress>0</defaultaddress>
```


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```
  <flagvat>0</flagvat>
  <notes>abc</notes>
  <delivery_notes>abc</delivery_notes>
</delivery>
```

Sample Result

```
<status>
  <code>0</code>
  <message>123</message>
</status>
```

Upon successful submission, the web service will return a status message containing:

Code = 0
Message = the generated/edited delivery address ID

SetOrder

Available in Merlin v100 onwards.

The `SetOrder` function provides the ability to generate a new sales order within the Merlin system. An order can be populated with any data provided in the submitted xml allowing you to specify custom invoice, delivery, credit card and price details. Once into Merlin the order goes through the standard order entry procedure, ensuring that all the order details are verified and the required account/stock checks are passed.

If the `<card_authamount>` element contains a decimal value greater than zero, then a deposit invoice will be created for the specified amount. This would then be redeemed as a deposit from within Merlin using the standard deposit procedures.

Parameters

Parameter	Type	Notes
datasource	string	connection string provided by Merlin
xml	string	validated against SetOrder.xsd

XML Parameter XSD

A link to the XSD for the XML parameter of the `SetOrder` function can be found at:

<<YOUR-WEB-SERVICE-URL>>/Schemas/SetOrder.xsd

Sample XML Parameter

```
<salesorder>
  <company>1</company>
  <depot>01</depot>
  <inv_account>abc</inv_account>
  <inv_name>abc</inv_name>
  <inv_add1>abc</inv_add1>
  <inv_add2>abc</inv_add2>
  <inv_city>abc</inv_city>
  <inv_county>abc</inv_county>
  <inv_country>abc</inv_country>
  <inv_postcode>abc</inv_postcode>
  <del_account>abc</del_account>
```


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```
<del_name>abc</del_name>
<del_add1>abc</del_add1>
<del_add2>abc</del_add2>
<del_city>abc</del_city>
<del_county>abc</del_county>
<del_country>abc</del_country>
<del_postcode>abc</del_postcode>
<card_name>abc</card_name>
<card_add1>abc</card_add1>
<card_add2>abc</card_add2>
<card_city>abc</card_city>
<card_county>abc</card_county>
<card_country>abc</card_country>
<card_postcode>abc</card_postcode>
<card_type>abc</card_type>
<card_no>abc</card_no>
<card_security>1</card_security>
<card_issue>1</card_issue>
<card_start>1</card_start>
<card_expire>1</card_expire>
<card_authamount>11.11</card_authamount>
<due_date>2000-01-01</due_date>
<ref>abc</ref>
<carriage>11.11</carriage>
<delivery>abc</delivery>
<notes>abc</notes>
<webref1>abc</webref1>
<webref2>abc</webref2>
<webref3>abc</webref3>
<contract>N</contract>
<salesman>1</salesman>
<contactname>abc</contactname>
<deliverycode>1</deliverycode>
<discount>1</discount>
<quote>Y</quote>
<despatch_layout>30</despatch_layout>
<item>
  <part>1</part>
  <depot>01</depot>
  <quantity>1</quantity>
  <price>11.11</price>
</item>
<item>
  <stockid>1</stockid>
  <desc>abc</desc>
  <desc2>abc</desc2>
  <desc3>abc</desc3>
  <desc4>abc</desc4>
  <desc5>abc</desc5>
  <quantity>1</quantity>
  <price>11.11</price>
  <disc1>11.11</disc1>
  <disc2>11.11</disc2>
  <disc3>11.11</disc3>
  <vat_inc>Y</vat_inc>
  <vat_code>1</vat_code>
  <reference>abc</reference>
  <unitconvid>0</unitconvid>
  <stkunitdescid>0</stkunitdescid>
  <pseudo>abc</pseudo>
```


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```
    <uoi>1</uoi>
    <duoi>each</duoi>
    <conversion_factor>11.11</conversion_factor>
    <bin>A1</bin>
    <certlevel>EASAF1</certlevel>
  </item>
</salesorder>
```

Sample Result

```
<status>
  <code>0</code>
  <message>S100123</message>
</status>
```

Upon successful processing of an order, the web service will return a status message containing:

Code = 0
Message = *the generated sales order number*

SetPOD

Available in Merlin v146 onwards.

The `SetPOD` function is used to add a POD document to a Sales Order.

The POD image file needs to be encoded using Base64 encoding.

Parameters

Name	Type	Notes
datasource	string	Connection string provided by Merlin
xml	xml	Validated against SetPOD.xsd

XML Parameter XSD

A link to the XSD for the XML parameter of the `SetPOD` function can be found at:

<<YOUR-WEB-SERVICE-URL>>/Schemas/SetPOD.xsd

Sample XML Parameter

```
<POD>
  <company>1</company>
  <depot>01</depot>
  <image>base64imagedata</image>
  <size>123</size>
  <inv_account>ABC</inv_account>
  <sales_order>S100123</sales_order>
</POD>
```

Sample Result

```
<status>
  <code>0</code>
```


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```
  <message>123</message>
</status>
```

Upon successful submission, the web service will return a status message containing:

Code = 0
Message = the generated POD ID

SetProspectContact

Available in Merlin v130 onwards.

The `SetProspectContact` function is used to either create or edit a prospect contact within Merlin. If the function is submitted with an XML containing an empty or zero `<keyfield>` element, then a new contact will be created for the prospect with the specified `<prospect_keyfield>`. If it is submitted with a value in the `<keyfield>` element, then the contact will be updated (if it exists within Merlin) or will be created using the specified account.

Parameters

Parameter	Type	Notes
datasource	string	Connection string provided by Merlin
xml	string	Validated against SetProspectContact.xsd

XML Parameter XSD

A link to the XSD for the XML parameter of the `SetProspectContact` function can be found at:

<<YOUR-WEB-SERVICE-URL>>/Schemas/SetProspectContact.xsd

Sample XML Parameter

```
<contact>
  <keyfield>0</keyfield>
  <company>1</company>
  <account>ABC</account>
  <prospect_keyfield>0</prospect_keyfield>
  <forename>ABC</forename>
  <middlename>ABC</middlename>
  <surname>ABC</surname>
  <salutation>ABC</salutation>
  <title>ABC</title>
  <add1>ABC</add1>
  <add2>ABC</add2>
  <city>ABC</city>
  <county>ABC</county>
  <country>ABC</country>
  <postcode>ABC</postcode>
  <office_phone>ABC</office_phone>
  <personal_phone>ABC</personal_phone>
  <mobile_phone>ABC</mobile_phone>
  <fax>ABC</fax>
  <sms>ABC</sms>
  <email>ABC</email>
  <memo>ABC</memo>
  <jobtitle>ABC</jobtitle>
  <accept_office>false</accept_office>
  <accept_personal>false</accept_personal>
  <accept_mobile>false</accept_mobile>
```


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```
<accept_fax>false</accept_fax>
<accept_sms>false</accept_sms>
<accept_email>false</accept_email>
<default_contact>0</default_contact>
<first_contact>1899-12-30 00:00</first_contact>
<next_contact>1899-12-30 00:00</next_contact>
<last_contact>1899-12-30 00:00</last_contact>
<frequency>0</frequency>
<default_filter>ABC</default_filter>
<default_calltime>1899-12-30 00:00</default_calltime>
<default_callday>0</default_callday>
<department>ABC</department>
<photograph>ABC</photograph>
<operator>0</operator>
<status>ABC</status>
<default_op_contact>0</default_op_contact>
<default_account_contact>0</default_account_contact>
<disable_callman>false</disable_callman>
<card_name>ABC</card_name>
<card_token>ABC</card_token>
<card_lastfour>ABC</card_lastfour>
<card_options>ABC</card_options>
<userchk1>false</userchk1>
<userchk2>false</userchk2>
<userchk3>false</userchk3>
<userchk4>false</userchk4>
<userchk5>false</userchk5>
<userchk6>false</userchk6>
<userchk7>false</userchk7>
<userchk8>false</userchk8>
<userchk9>false</userchk9>
<userchk10>false</userchk10>
<userchk11>false</userchk11>
<userchk12>false</userchk12>
<userchk13>false</userchk13>
<userchk14>false</userchk14>
<userchk15>false</userchk15>
<userchk16>false</userchk16>
<userchk17>false</userchk17>
<userchk18>false</userchk18>
<userchk19>false</userchk19>
<userchk20>false</userchk20>
<userchk21>false</userchk21>
<flag_override>0</flag_override>
<user1>ABC</user1>
<user2>ABC</user2>
<user3>ABC</user3>
<user4>ABC</user4>
<user5>ABC</user5>
<user6>ABC</user6>
<user7>ABC</user7>
<user8>ABC</user8>
<user9>ABC</user9>
<user10>ABC</user10>
<userc1>ABC</userc1>
<userc2>ABC</userc2>
<userc3>ABC</userc3>
<userc4>ABC</userc4>
<userc5>ABC</userc5>
<userc6>ABC</userc6>
```


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```
  <userc7>ABC</userc7>
  <userc8>ABC</userc8>
  <userc9>ABC</userc9>
  <userc10>ABC</userc10>
  <usern1>0</usern1>
  <usern2>0</usern2>
  <usern3>0</usern3>
  <usern4>0</usern4>
  <usern5>0</usern5>
  <usern6>0</usern6>
  <usern7>0</usern7>
  <usern8>0</usern8>
  <usern9>0</usern9>
  <usern10>0</usern10>
  <alert_type>0</alert_type>
  <personal_folder>ABC</personal_folder>
  <gdpr_templateid>0</gdpr_templateid>
</contact>
```

Sample Result

```
<status>
  <code>0</code>
  <message>123</message>
</status>
```

Upon successfully inserting a contact note, the web service will return a status message containing:

Code = 0
Message = *the generated prospect contact ID*

SetPurchaseOrder

Available in Merlin v148 onwards.

The `SetPurchaseOrder` function provides the ability to generate a new purchase order within the Merlin system. An order can be populated with any data provided in the submitted xml allowing you to specify custom invoice, delivery, credit card and price details. Once into Merlin the order goes through the standard purchase order entry procedure, ensuring that all the order details are verified and the required account/stock checks are passed.

Parameters

Parameter	Type	Notes
datasource	string	connection string provided by Merlin
xml	string	validated against SetPurchaseOrder.xsd

XML Parameter XSD

A link to the XSD for the XML parameter of the `SetPurchaseOrder` function can be found at:

<<YOUR-WEB-SERVICE-URL>>/Schemas/SetPurchaseOrder.xsd

Sample XML Parameter

```
<purchaseorder>
  <company>1</company>
  <depot>01</depot>
  <inv_account>ABBEY</inv_account>
```


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```
  <inv_name>abc</inv_name>
  <inv_add1>abc</inv_add1>
  <inv_add2>abc</inv_add2>
  <inv_city>abc</inv_city>
  <inv_county>abc</inv_county>
  <inv_country>abc</inv_country>
  <inv_postcode>abc</inv_postcode>
  <due_date>2018-01-01</due_date>
  <ref>abc</ref>
  <supplier_ref>abc</supplier_ref>
  <memo>abc</memo>
  <carriage>11.11</carriage>
  <delivery_notes>abc</delivery_notes>
  <order_notes>abc</order_notes>
  <contract>N</contract>
  <buyer>1</buyer>
  <contact_name>abc</contact_name>
  <delivery_code>1</delivery_code>
  <discount>1</discount>
  <quote>Y</quote>
  <flag_direct>N</flag_direct>
  <contract_order>N</contract_order>
  <cofc>Y</cofc>
  <item>
    <part>B001</part>
    <depot>01</depot>
    <quantity>1</quantity>
    <price>11.11</price>
  </item>
  <item>
    <message>xyz</message>
  </item>
  <item>
    <stockid>30</stockid>
    <desc>abc</desc>
    <desc2>abc</desc2>
    <desc3>abc</desc3>
    <desc4>abc</desc4>
    <desc5>abc</desc5>
    <quantity>1</quantity>
    <price>11.11</price>
    <disc1>11.11</disc1>
    <disc2>11.11</disc2>
    <disc3>11.11</disc3>
    <disc4>11.11</disc4>
    <vat_inc>N</vat_inc>
    <vat_code>2</vat_code>
    <reference>abc</reference>
    <mpn>abc</mpn>
    <bin>A1</bin>
    <cofc>Y</cofc>
    <uoi>1</uoi>
    <duoi>box</duoi>
    <conversion_factor>5</conversion_factor>
  </item>
</purchaseorder>
```


Sample Result

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```
<status>
  <code>0</code>
  <message>P100123</message>
</status>
```

Upon successful processing of an order, the web service will return a status message containing:

Code = 0
Message = *the generated purchase order number*

SetAppointment

The `SetAppointment` function is used to create or amend an appointment within Merlin. If the function is submitted with an XML containing an empty or zero `<appointmentid>` element, then a new appointment will be created for the contact with the specified `<contactid>`.

Parameters

Parameter	Type	Notes
datasource	string	Connection string provided by Merlin
xml	string	Validated against SetAppointment.xsd

XML Parameter XSD

A link to the XSD for the XML parameter of the `SetAppointment` function can be found at:

<<YOUR-WEB-SERVICE-URL>>/Schemas/SetAppointment.xsd

Sample XML Parameter

```
<appointment>
    <company>1</company>
    <appointmentid>0</appointmentid>
    <contactid>123</contactid>
    <source>C</source>
    <date_time>1899-12-30 00:00</date_time>
    <operator>0</operator>
    <date_created>1899-12-30 00:00</date_created>
    <operator_created>1</operator_created>
    <type>0</type>
    <description>abc</description>
    <reference>abc</reference>
    <priority>0</priority>
    <date_reminder>1899-12-30 00:00</date_reminder>
    <duration>0</duration>
    <status>0</status>
    <linkid>0</linkid>
    <location>0</location>
    <campaignid>0</campaignid>
    <leadid>0</leadid>
    <action>0</action>
</appointment>
```

SetSql

Available in Merlin v100 onwards.

The `SetSql` function is for internal use only.

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SetSqlWithKey

Available in Merlin v137 onwards.

The `SetSqlWithKey` function is for internal use only.

ValidateFeature

Available in Merlin v129 onwards.

The `ValidateFeature` function provides the ability to check if a feature is enabled within Merlin.

Parameters

Parameter	Type	Notes
datasource	string	Connection string provided by Merlin
company	integer	A valid company number
code	string	Merlin feature code

Sample Result

```
<status>
  <code>54</code>
  <message>feature not enabled</message>
</status>
```

If the feature is enabled, it will return:

Code = 0
Message = *Ok*

If the feature is not enabled, it will return:

Code = 54
Message = *feature not enabled*

Status Messages

Whenever data is submitted to the web service, you will receive an xml reply containing the data you requested or a status message.

Sample Result

```
<status>
  <code>31</code>
  <message>customer not found</message>
</status>
```

The following status messages can be returned:

Code	Details
0	OK
10	Database connection error
11	Invalid SQL
12	Invalid XML
13	No Records inserted

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14	Database version incorrect - contact merlin business software
15	Exception from SQL
21	Company not found
22	Depot not found
31	Customer not found
32	Transaction not found
33	Delivery Address not found
34	Invalid VAT Code
35	Invalid Country Code
36	Invalid Delivery Code
37	Invalid Price Code
38	Invalid User Field
51	Stock not found
52	Stock price not found
53	Delivery Code not found
54	Feature Not Enabled
55	Temporarily Unavailable (Period End being run)
56	Invalid Key
57	Customer on Stop
61	Supplier not found
62	Purchase Order not found
63	Purchase Order Line not found
71	Invalid Channel Config ID
72	Fatal error occurred
73	Nominal not found
74	Cannot change Currency while Transactions/Orders exist
75	Cannot change Currency while Discounts exist
76	Parameters missing
77	Parameter count mismatch
81	Invalid event type
82	Invalid parameter value
-1	Unknown error

Any code other than zero shows that the request has not been successfully completed. In this case, the message would show a value associated with the particular function (i.e. sales order number, customer account, etc.).

File Transfer

If the web service is undesirable or unavailable for a particular setup, a file transfer option is also available. Routines can be put in place to export data into xml files that can then be transferred to another server/pc. This can be either an automated routine (that creates and imports files at regular intervals) or a manual process (that is triggered when required).

Stock

The Stock routine can output a complete stock list as an xml file (named *stock.xml*), the format of which is shown in the sample below. Stock can be flagged as available for use on a website by using the `web_available` flag held against each stock record. The routine will output only the relevant records into a standard format containing commonly used details – see example below.

Sample Result

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```
<data count="1" count_available="9">
  <row id="1">
    <stockid>9</stockid>
    <company>9</company>
    <depot>abc</depot>
    <part>abc</part>
    <qty_free>9</qty_free>
    <desc1>abc</desc1>
    <desc2>abc</desc2>
    <desc3>abc</desc3>
    <desc4>abc</desc4>
    <desc5>abc</desc5>
    <list_price>9.99</list_price>
    <uoi>9</uoi>
    <duoi>abc</duoi>
    <web_image>abc</web_image>
    <wines_flag>9</wines_flag>
    <wines_abv>9.99</wines_abv>
    <wines_vintage>9</wines_vintage>
    <wines_region>abc</wines_region>
    <wines_volume>9.99</wines_volume>
  </row>
</data>
```

Orders

The `SetOrder` function is used to create a new order within the Merlin system. The order will be populated with any data provided allowing you to specify custom invoice, delivery, credit card, and price details. Once into Merlin the order goes through the standard order entry procedure, ensuring that nothing is overlooked.

Each order that you want to transfer back into Merlin must be specified in a single file xml that is validated against the XSD file (shown in Figure 1 - SetOrder XSD). The name of the files is irrelevant, but they must be located in a single directory that is accessible by the Merlin server.

Sample XML

```
<salesorder>
  <company>1</company>
  <depot>01</depot>
  <inv_account>abc</inv_account>
  <inv_name>abc</inv_name>
  <inv_add1>abc</inv_add1>
  <inv_add2>abc</inv_add2>
  <inv_city>abc</inv_city>
  <inv_county>abc</inv_county>
  <inv_country>abc</inv_country>
  <inv_postcode>abc</inv_postcode>
  <del_account>abc</del_account>
  <del_name>abc</del_name>
  <del_add1>abc</del_add1>
  <del_add2>abc</del_add2>
  <del_city>abc</del_city>
  <del_county>abc</del_county>
  <del_country>abc</del_country>
  <del_postcode>abc</del_postcode>
  <card_name>abc</card_name>
  <card_add1>abc</card_add1>
  <card_add2>abc</card_add2>
```

```
  <card_city>abc</card_city>
  <card_county>abc</card_county>
  <card_country>abc</card_country>
  <card_postcode>abc</card_postcode>
  <card_type>abc</card_type>
  <card_no>abc</card_no>
  <card_security>1</card_security>
  <card_issue>1</card_issue>
  <card_start>1</card_start>
  <card_expire>1</card_expire>
  <card_authamount>11.11</card_authamount>
  <due_date>2000-01-01</due_date>
  <ref>abc</ref>
  <carriage>11.11</carriage>
  <delivery>abc</delivery>
  <notes>abc</notes>
  <webref1>abc</webref1>
  <webref2>abc</webref2>
  <webref3>abc</webref3>
  <item>
    <part>1</part>
    <depot>01</depot>
    <quantity>1</quantity>
    <price>11.11</price>
  </item>
  <item>
    <stockid>1</stockid>
    <desc>abc</desc>
    <desc2>abc</desc2>
    <desc3>abc</desc3>
    <desc4>abc</desc4>
    <desc5>abc</desc5>
    <quantity>1</quantity>
    <price>11.11</price>
    <disc1>11.11</disc1>
    <disc2>11.11</disc2>
    <disc3>11.11</disc3>
    <vat_inc>Y</vat_inc>
    <vat_code>1</vat_code>
  </item>
</salesorder>
```

Sample Result

```
<status>
  <code>0</code>
  <message>123456</message>
</status>
```

Upon successful processing of an order, the web service will return a status message containing:

Code = 0
Message = *the generated sales order number*

Resources

Further Information

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If you require more information on the Merlin Web Module, then please log a Support case via AppCentral or telephone +44 (0)1246 457170

Test Site

All installations of the Merlin Web Module have both a production (or live) and development (or training) sites. Usually these are configured on two different ports, i.e.

- Production – port 8181
- Development – port 8182

All testing is done on the client's development system.

Online Schema Validators

There are several online XSD schema validators that can be used to check your submitted XML files are valid:

- http://www.corefiling.com/opensource/schemaValidate.html
- http://www.utilities-online.info/xsdvalidation/

Sample SQL Queries

Basic Stock Information

Query all stock parts and descriptions, along with base prices and quantity breaks, that exist in company 1 depot 01:

```
SELECT a.company, a.depot, a.part, a.desc1, a.desc2, a.desc3, a.desc4,
  a.desc5, a.qty_hand, b.price, b.price1, b.break1, b.price2, b.break2,
  b.price3, b.break3, b.price4, b.break4, b.price5, b.break5
FROM stock AS a
JOIN stprice AS b
ON a.stockid = b.stockid
WHERE a.company = 1
AND a.depot = '01';
```

Pictures

Query pictures that have been amended in the last 7 days:

```
SELECT picture.company, picture.depot, picture.part,
  ENCODE( picture.picture, 'base64' ) AS picture, picture.date_amended
FROM picture
WHERE picture.source = 0
AND picture.date_amended > CURRENT_DATE - '7 days'::INTERVAL;
```

HTML Fields

Example to return a TEXT field that contains HTML. PostgreSQL currently does not like DOCTYPE so these need to be removed before casting the field to XML.

```
SELECT XPATH('/HTML/BODY/child::*',replace(replace(usr_description,'<!DOCTYPE HTML
PUBLIC "-//W3C//DTD HTML 4.0 Transitional//EN">',''),'<!DOCTYPE html PUBLIC "-
//W3C//DTD HTML 4.0 Transitional//EN" "">',''))::xml)
FROM webstock
```

Sample PHP Code

GetSql using SOAP

The example demonstrates the GetSQL function, using a SOAP request.

```
// web service url
$url = 'http://<<YOUR-WEB-SERVICE-URL>>:8181/merlin.asmx?WSDL';

// data
$data = array();
$data['datasource'] = '1c4cb0f4eb';
$data['query'] = 'SELECT customer.name, customer.add1, customer.postcode' .
    ' FROM customer' .
    ' WHERE customer.company = 1' .
    ' AND customer.account = \'ABC123\'' .
    ' LIMIT 1';

// create soap client
$merlin = new SoapClient( $url );

// execute GetSQL and return response
$result = $merlin->GetSql( $data )->GetSqlResult;

// dump result
print_r( $result );
```

SetOrder using Curl

This example demonstrates the SetOrder function using Curl.

```
// web service url
$host = '<<YOUR-WEB-SERVICE-URL>>';
$page = '/Merlin.asmx';
$url = 'http://' . $host . ':8181' . $page . '?WSDL';

// data
$datasource = '1c4cb0f4eb';
$xml = '<salesorder>' .
    '<company>1</company>' .
    '<depot>01</depot>' .
    '<inv_account>BOL14789</inv_account>' .
    '<due_date>2015-04-17</due_date>' .
    '<carriage>5.95</carriage>' .
    '<item>' .
    '  <part>B70011</part>' .
    '  <quantity>1</quantity>' .
    '</item>' .
    '</salesorder>';

$soapEnvelope = '<soap:Envelope xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">' .

    '  <soap:Body>' .
    '    <SetOrder xmlns="CCBS">' .
    '      <datasource>' . htmlspecialchars( $datasource ) . '</datasource>' .
    '      <xml>' . htmlspecialchars( $xml ) .'</xml>' .
    '    </SetOrder>' .
    '  </soap:Body>' .
    '</soap:Envelope>';
```


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```
// create curl client
$curl = curl_init();
curl_setopt( $curl, CURLOPT_USERAGENT, 'Mozilla/5.0 (X11; U; FreeBSD i386; en-US;
rv:1.2a) Gecko/20021021' );
curl_setopt( $curl, CURLOPT_RETURNTRANSFER, 1 );
curl_setopt( $curl, CURLOPT_SSL_VERIFYPEER, FALSE );
curl_setopt( $curl, CURLOPT_TIMEOUT, 180 );
curl_setopt( $curl, CURLOPT_URL, $url );
curl_setopt( $curl, CURLOPT_POSTFIELDS, $soapEnvelope );
curl_setopt( $curl, CURLOPT_HTTPHEADER, array(
        'POST ' . $page . ' HTTP/1.1',
        'HOST: ' . $host,
        'Content-Type: text/xml; charset=utf-8',
        'Content-length: ' . strlen( $soapEnvelope ),
        'SOAPAction: "CCBS/SetOrder"',
    ) );

// execute SetOrder and return response
$result = curl_exec( $curl );

// dump result
print_r( $result );
```


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